

Static / Program Analysis Engineer: C#.NET static analysis and symbolic-computation/symbolic-execution tasks at ISP RAS; previously LLVM 22/C++ at MCST. Public proof points: Roslyn CFG fixed-point data flow and interprocedural LLVM analysis with conservative applicability checks.

2022: Compilers / VMs · 2023: CIL + native x86-64 · 2024–2025: AST / IR / backends → toolchain · 2026 → : Professional LLVM + static analysis

PROFESSIONAL EXPERIENCE	SKILLS
2026 — present ISP RAS — Static Analysis Engineer - Contribute to SharpChecker, ISP RAS's C#.NET static-analysis platform. - Current work includes symbolic-computation and symbolic-execution tasks in C#.NET program analysis. July — August 2026 MCST — Compiler Engineering Intern - Implemented a LICM pass for LLVM 22 in C++; hoisting conditions check loop structure, memory access, side effects, and speculative safety. - Built a Python harness comparing program behavior before/after optimization and checking transform/no-transform cases.	Data-flow Analysis Roslyn CFG, worklist/fixed point, edge refinement, predecessor joins, loops/back edges Interprocedural / LLVM effects analysis, affine evolution, APInt, transformation applicability, LLVM IR Analysis Correctness assignment/ref/out invalidation, conservative unsupported-case handling, regression tests, LLVM Verifier Languages / Tools C#, C++23, Roslyn, LLVM 22, .NET
PROJECTS	EDUCATION
DerefAfterNullAnalyzer A Roslyn ControlFlowGraph analyzer propagates null state on conditional edges to a fixed point, joins predecessor states, and reprocesses successors for loops/back edges. Method/property/field/index/event dereferences; invalidation after assignment/ref/out; dedicated analyzer tests use Microsoft.CodeAnalysis.CSharp.Testing. LLVM Interprocedural Global IV Optimization An LLVM 22 pass localizes supported induction-like globals after interprocedural effect analysis, affine evolution, and a separate applicability check; APInt models fixed-width wraparound. 29 positive/negative regression cases cover LLVM Verifier checks, structural expectations, idempotence, and observable before/after behavior.	HSE University — Software Engineering, 2026–2030
	RECOGNITION
	LangDev'26 — Build the Language, Then Make the Abstractions Disappear Accepted talk on UniversalToolchain's architecture; LangDev'26 takes place October 8–9, 2026 in Málaga. UniversalToolchain — Baltic Science and Engineering Competition First Degree Diploma and Grand Prize "Perfection as Hope" for UniversalToolchain. MEPhI Junior Overall winner; First Degree Diploma, 96/100 for UniversalToolchain. Moscow / remote